

CapNav: Benchmarking Vision Language Models on Capability-conditioned Indoor Navigation

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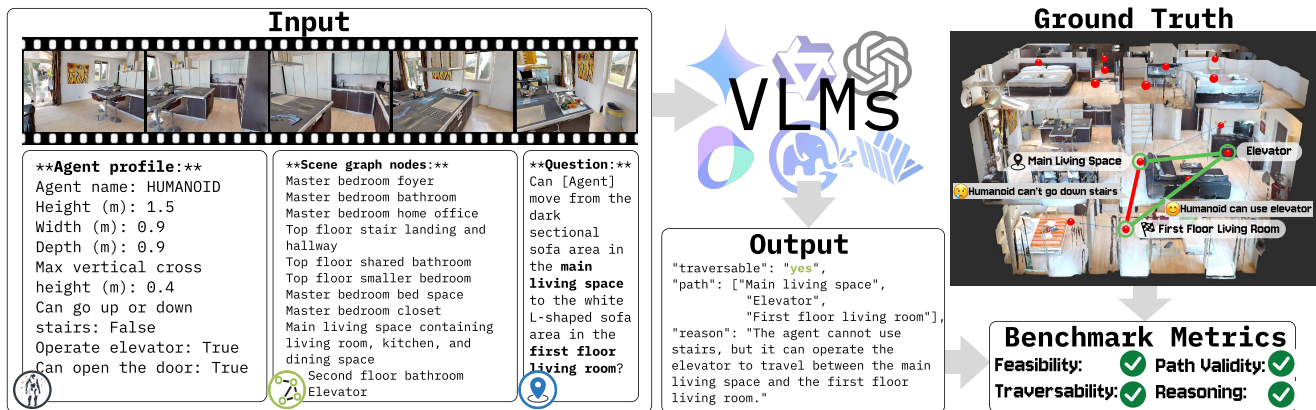


Figure 1. We introduce Capability-Conditioned Navigation (**CapNav**), a benchmark designed to evaluate how well VLMs can navigate complex indoor spaces given an agent’s specific physical and operational capabilities. CapNav inputs (1) a tour video of an indoor space, (2) nodes of its navigation graph, (3) an agent’s mobility profile, and (4) a navigation task, and evaluates VLM outputs in task feasibility, path validity, route traversability, and reasoning validity.

Abstract

Vision-Language Models (VLMs) have shown remarkable progress in Vision-Language Navigation (VLN), offering new possibilities for navigation decision-making that could benefit both robotic platforms and human users. However, real-world navigation is inherently conditioned by the agent’s mobility constraints. For example, a sweeping robot cannot traverse stairs, while a quadruped can. We introduce **Capability-Conditioned Navigation (CapNav)**, a benchmark designed to evaluate how well VLMs can navigate complex indoor spaces given an agent’s specific physical and operational capabilities. CapNav defines five representative human and robot agents, each described with physical dimensions, mobility capabilities, and environmental interaction abilities. CapNav provides 45 real-world indoor scenes, 473 navigation tasks, and 2365 QA pairs to test if VLMs can traverse indoor environments

based on agent capabilities. We evaluate 13 modern VLMs and find that current VLM’s navigation performance drops sharply as mobility constraints tighten, and that even state-of-the-art models struggle with obstacle types that require reasoning on spatial dimensions. We conclude by discussing the implications for capability-aware navigation and the opportunities for advancing embodied spatial reasoning in future VLMs. The benchmark is available at <https://makeabilitylab.github.io/CapNav/>

1. Introduction

As vision-language models (VLMs) advance in visual grounding [47] and spatial reasoning [11], they are now commonly used as drop-in “navigation assistants” or planning modules for robots and assistive systems [16, 46, 66]. Guiding people to reach destinations [16, 30] and controlling robots in movement decisions [20], they are widely applied in open-ended scenarios where solutions can be plu-

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2. Related Work

2.1. Vision Language Navigation and Benchmarks

Vision-Language Navigation (VLN) studies how an agent follows natural-language instructions using grounded visual observations to traverse an environment and reach a goal [2, 67]. Since the introduction of Room-to-Room (R2R) [2], which casts indoor navigation as step-by-step action selection on a sparse connectivity graph, a wide range of datasets and benchmarks have expanded VLN along multiple axes. Indoor instruction-following has been scaled in path length, diversity, and grounding density (e.g., R4R [22] and RxR [26]); broader urban/street settings have also been explored (e.g., TOUCHDOWN [9], StreetLearn [35], CityLearn [7], CityWalker [33], Talk2Nav [53]). Other benchmarks couple navigation with object grounding or task execution (e.g., REVERIE [39], ALFRED [48], ObjectNav [3], EmbodiedBench [64]), while newer efforts probe long-horizon and open-ended embodied behaviors (e.g., GOAT-bench [24] and NavBench [40]).

Beyond task variants, VLN benchmarks differ in their observation and interaction assumptions. The canonical setting is *interactive*: an agent explores incrementally, receiving egocentric observations and accumulating spatial knowledge step-by-step. In contrast, a *passive* formulation provides global context, such as panoramic scans or pre-recorded videos, aiming to evaluate high-level spatial reasoning while reducing confounds from exploration and low-level control. For example, VideoNavQA [5] replaces interactive navigation with oracle trajectory videos to isolate embodied visual reasoning; NRNS [18] learns image-goal navigation from passive roaming videos without online RL or a simulator; and PONI [42] trains an object-search module interaction-free from offline semantic maps. OpenEQA [34] further supports an episodic-memory (video) setting that evaluates spatial memory and reasoning over recorded tours and scans. Under such passive formulations, the task increasingly resembles route planning and feasibility assessment; following prior work [5, 34], we still refer to this setting as navigation.

Our goal—benchmarking VLM navigation validity under explicit mobility constraints—benefits from global observation, since it allows models to reason about traversability and alternative routes across the entire scene while isolating embodiment constraints from exploration and execution noise. We further adopt a graph-based spatial abstraction [2, 26] to support pluralistic solutions and enable fine-grained evaluation of path feasibility and validity. This complements existing VLN benchmarks by enabling structured, capability-conditioned comparisons across real-world environments and embodiments.

2.2. Navigation with mobility constraints

Past navigation research has modeled a wide range of mobility constraints, including accessibility for wheelchair users [15, 43, 65], assistance for people who are blind or low vision [13, 36, 55], and robotics embodiments such as quadrupeds [19, 27, 32], humanoids [10], and vehicles operating over varied terrain [23, 29]. Some works also consider settings where multiple mobility profiles co-exist, either through simulator diversity [58–60] or by modeling user preferences [28]. Collectively, these studies highlight that embodiment constraints materially change feasible routes and navigation strategies.

However, most evaluations remain siloed—focusing on a single embodiment, environment family, or constraint type—making cross-embodiment comparison difficult on matched tasks. Recent efforts begin to address this gap: VLN-PE (introduced with a holistic study of physical and visual disparities) enables systematic cross-embodiment evaluation in a physically grounded simulator setting [54]; NaviTrace [56] evaluates embodiment-conditioned navigation via 2D trace prediction in real-world images; and VAMOS [6] introduces an affordance-grounded hierarchical planner to modulate plans by embodiment capabilities. Nonetheless, existing benchmarks either emphasize simulator-centric embodiment effects or cast the problem as trace/plan prediction without explicitly validating pluralistic, capability-conditioned route feasibility in complex indoor spaces with realistic navigation obstacles (e.g., multi-floor transitions, bottlenecks, and accessibility-specific affordances).

3. The CapNav Benchmark

We introduce *Capability-conditioned Navigation* (CapNav), a benchmark for assessing VLMs’ ability to navigate built environments under specific physical capabilities.

3.1. Problem Statement

In the CapNav task, each query instance is defined by a *Space–Task–Capability* triple $\langle \mathcal{S}, \tau, \mathbf{a} \rangle$ where the space $\mathcal{S}$ is represented by a touring video and a set of key spatial nodes; τ is a natural-language navigation goal specifying the source and target locations; and $\mathbf{a}$ is an agent profile encoding physical dimensions and operational abilities. Given $\langle \mathcal{S}, \tau, \mathbf{a} \rangle$, a vision language model produces

$$(\hat{y}, \hat{P}, \hat{\rho}) = f_{\theta}(\mathcal{S}, \tau, \mathbf{a}),$$

where $\hat{y} \in \{0, 1\}$ denotes task feasibility (CAN / CANNOT complete), $\hat{P} = [v_0, \dots, v_m]$ is a node sequence representing the proposed navigation path, and $\hat{\rho}$ is a concise rationale explaining infeasibility when $\hat{y} = 0$.

This formulation enables CapNav to holistically evaluate a VLM’s capability-aware navigation ability. We assess

performance from four complementary aspects: (i) *navigation feasibility*, (ii) *path validity*, (iii) *traversability accuracy*, and (iv) *reasoning quality* for infeasible cases.

Space. Each space $\mathcal{S}$ is a scanned 3D indoor environment comprising multiple interconnected rooms. We provide spatial information as a rendered touring video. To ground the navigation to language description, we manually create a connectivity graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ over spatial nodes (e.g., *kitchen*, *entry foyer*, *hallway*) and edges that denote directly walkable connections.

Task. A navigation task τ is a natural-language instruction specifying a movement goal from a source node to a target node (e.g., “*From the wooden cabinet in the entrance foyer, go to the desk area in the master bedroom.*”). Formally, each goal τ can be expressed as a directed pair:

$$\tau : (v_{\text{src}}, d_{\text{src}}) \rightarrow (v_{\text{tgt}}, d_{\text{tgt}}), \quad v_{\text{src}}, v_{\text{tgt}} \in \mathcal{V},$$

where v_{src} and v_{tgt} denote the source and target nodes in $\mathcal{V}$, and d_{src} and d_{tgt} are their corresponding natural-language descriptions specifying the precise starting and ending positions within those nodes (e.g., “*at the wooden cabinet in the entrance foyer*”, “*beside the desk area in the master bedroom*”). This representation captures both the high-level spatial nodes and the finer-grained textual localization needed for realistic indoor navigation reasoning.

Agent Profiles We specify the mobility capabilities $\mathbf{a}$ with five distinct but representative agent profiles that cover human mobility and common robot platforms. **Adult with no motor disabilities** represent a default condition where all indoor routes can be achieved; **Wheelchair users** cannot use stairs and require clearance for passing and turning; **Humanoid robot** cannot go up/down stairs and require clearance for passing; **Sweeping robots** require a flat floor surface; **Quadrupedal robots** can traverse most indoor spaces but cannot operate doors/elevators.

Each profile is described by a capability json $\mathbf{a} = (\phi, \kappa, \mu)$, where ϕ captures the physical footprint; κ captures vertical traversal limits, including vertical height of a single obstacle, and whether the agent can cross continuous stairs; and μ captures operation/manipulation abilities. These attributes directly affect traversability on edges $\mathcal{E}$ and on overall tasks τ . See Figure 4 for examples.

3.2. Ground Truth of Traversability

Traversability is manually annotated per-task and per-agent type. They are binary labels at the *edge* level, allowing multiple possible routes and providing more precise reasoning if a route is not traversable. For each navigation task τ

and embodiment $\mathbf{a}$, annotators iterate through each possible simple path that connects v_{src} and v_{tgt} in $\mathcal{G}$, and assign $g_e^{(\mathbf{a})} \in \{0, 1\}$, to each edge e on these paths, indicating whether e is passable under $\mathbf{a}$ given local spatial geometry and mobility capabilities. Reasons for non-traversability are annotated as text descriptions (e.g., “*cannot go up/down stairs*”). The annotation UI visualizes 3D colliders matching ϕ and supports manually controlled moving/turning to verify clearance and turning space (Fig. 3).

Given per-edge labels, we can then define the feasibility ground truth for a task (s, g) as the existence of at least one simple path (an acyclic sequence of distinct nodes) $P(v_{\text{src}}, v_{\text{tgt}})$ that connects v_{src} and v_{tgt} in $\mathcal{G}$, where all edges are traversable:

$$y^* = \mathbb{I}[\exists P(v_{\text{src}}, v_{\text{tgt}}) : \forall (u, v) \in P, g_{(u,v)}^{(\mathbf{a})} = 1],$$

3.3. Dataset

We collect 3D indoor scenes from HM3D [41] and Matterport3D [8], and record touring videos in the Habitat simulator [44] by manually navigating each space. Videos are rendered at 2FPS from human-eye height (1.5m) with a 75 field of view to mimic casual handheld walkthroughs. For each scene, we construct a navigation graph via a custom annotation interface: nodes $v \in \mathcal{V}$ are annotated with semantic labels $c(v)$ and approximate 3D positions, and edges $e = (u, v) \in \mathcal{E}$ denote bidirectional, directly walkable proximity. We provide each scene’s video and node list to Gemini 2.5 Pro to generate navigation tasks, manually verify task validity, and annotate edge-level traversability for five embodiments, including reasons for non-traversable cases. After filtering out incomplete scenes, disconnected subspaces, repetitive semantics, or trivial layouts, we retained 45 indoor scenes (avg. 160.38 s per video, 13.8 nodes, 14.5 edges), yielding 2,365 navigation tasks and 5,075 traversability labels (3,945 positive, 1,130 negative). Feasibility varies across agent types on both task-level and edge-level (Table 1).

Table 1. Ratio of feasible task and traversable edges per agent type

Agent Type	Feasible Task Ratio	Edge Traversable Ratio
Human	1	1
Wheelchair	0.48	0.71
Humanoid	0.22	0.43
Quadrupedal	0.97	0.96
Sweeper	0.57	0.79

3.4. Metrics & Scoring

We evaluate model outputs against ground truth using four complementary metrics, reported both overall and per-embodiment. Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be the space graph, let

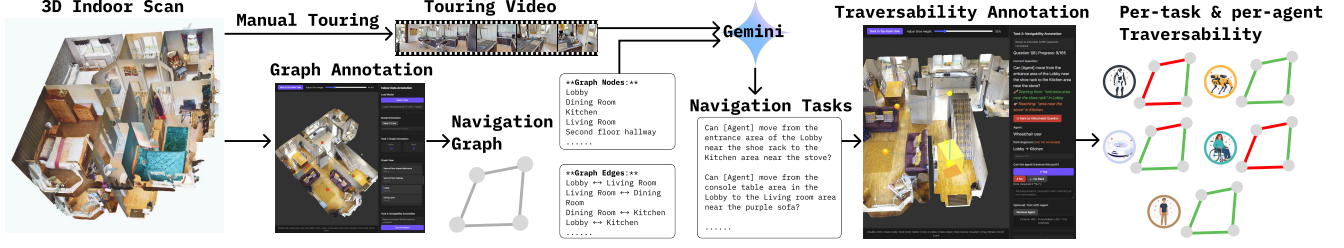


Figure 3. Overview of CapNav’s data construction: Starting from a 3D indoor scan, we manually record a touring video and a navigation graph. We then use Gemini to generate natural language navigation tasks. Finally, per-task and per-agent traversability are annotated by manually controlling agents in the annotation interface.

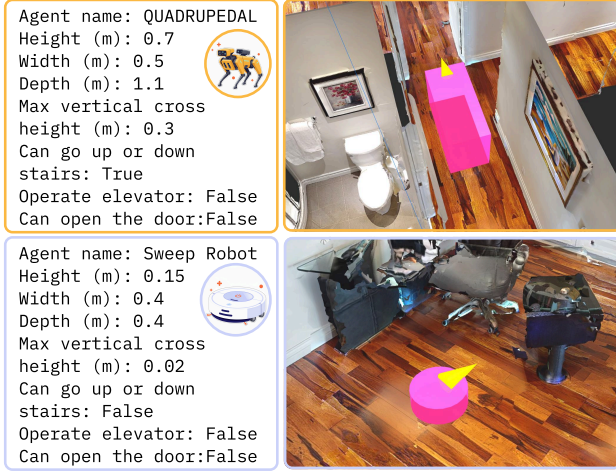


Figure 4. Examples of CapNav’s agent profiles. Left: each profile specifies the physical dimensions and functional capabilities. Right: the agents’ physical dimensions are rendered and maneuvered in the 3D scenes to help confirm traversability.

$P(v_{\text{src}}, v_{\text{tgt}})$ denote a simple path in $\mathcal{G}$, and let $E(\hat{P}) \subseteq \mathcal{E}$ be the ordered edge set of a predicted path $\hat{P}$.

(1) Feasibility Classification (Feas-F1). We report the F_1 score for binary feasibility prediction.

(2) Path Validity (PV). A predicted route $\hat{P} = [v_0, \dots, v_m]$ is considered valid if (i) $v_i \in \mathcal{V}$ for all i , (ii) $(v_i, v_{i+1}) \in \mathcal{E}$ for all i , (iii) $v_0 = v_{\text{src}}$ and $v_m = v_{\text{tgt}}$, and (iv) $\hat{P}$ is a simple path (no node repetitions). We report:

$$\text{PV} = \mathbb{E}[\mathbb{I}(\hat{P} \in \mathcal{P}_{\text{simple}}(v_{\text{src}}, v_{\text{tgt}}))].$$

(3) Route Traversability Accuracy (RTA). Conditioned on $\hat{y} = 1$ and a valid $\hat{P}$, we evaluate the fraction of edges in $\hat{P}$ that are actually traversable under embodiment $\mathbf{a}$:

$$\text{RTA}(\hat{P}, \mathbf{a}) = \frac{\sum_{e \in E(\hat{P})} g_e^{(\mathbf{a})}}{|E(\hat{P})|},$$

where $g_e^{(\mathbf{a})} \in \{0, 1\}$ is the edge-level traversability label. All edges are assigned unit weight so each edge contributes equally. $\text{RTA} = 1$ indicates a fully valid route, while lower values quantify partial failures. This metric serves a role analogous to path-quality measures such as SPL [2], but conditions on embodiment-specific traversability.

(4) Reasoning Validity (RV). For infeasible predictions ($\hat{y} = 0$), the model must (i) return a proposed route $\hat{P}$ and (ii) provide a short rationale $\hat{\rho}$ explaining the failure. Let $\mathcal{R}^*(\hat{P}, \mathbf{a})$ be the set of annotated failure reasons, we define an LLM-as-judge function J_{LLM} that returns 1 if the model’s rationale semantically matches the annotation and 0 otherwise. We verified the reliability of this function by manually validating 300 randomly sampled VLM reasonings. The results show an 89% alignment between human and LLM verdicts, indicating effectiveness of the LLM-as-judge method. We report RV as the mean of binary judgments:

$$\text{RV} = \mathbb{E}[J_{\text{LLM}}(\hat{\rho}, \mathcal{R}^*(\hat{P}, \mathbf{a}))].$$

Composite CapNav Score. We report a composite score that aggregates the above aspects:

$$\text{CapNav} = \lambda_c F_1 + \lambda_p \text{PV} + \lambda_t \overline{\text{RTA}} + \lambda_r \overline{\text{RV}}, \sum \lambda = 1,$$

where $\overline{\text{RTA}}$ averages over positive predictions, and $\overline{\text{RV}}$ averages over negative predictions. We report per-embodiment scores and a macro-average across embodiments. By default, we set all λ as 0.25.

Reference Upper and Lower Bounds. We establish both lower and upper reference bounds under the proposed evaluation metrics. We simulate a random-walk policy on the navigation graph that ignores scene and agent constraints. This naive baseline achieves a lower-bound score of 29.35. As an approximate upper bound, we evaluate human performance on four participants, each completing 20 randomly sampled CapNav tasks, yielding CapNav scores ranging from 48.97 to 74.77 (AVG = 60.59).

Model	Feas-F1	PV	RTA	RV	CapNav	HUMAN	HUMANOID	QUAD	SWEEP	WHEEL
Non-thinking mode										
Intern-VL3.5-8B [68]	73.65	26.53	39.54	27.17	41.72	53.89	35.61	51.92	51.36	49.43
Intern-VL3.5-14B [68]	75.68	35.82	50.21	25.18	46.72	50.43	42.98	47.69	44.66	47.38
Keye-VL-1.5-8B [62]	77.63	30.84	43.96	36.70	47.28	58.64	53.44	54.37	62.17	60.11
MiMo-VL-7B-RL-2508 [50]	57.31	43.61	60.08	29.35	47.59	46.61	38.96	45.38	41.46	50.85
Qwen3-VL-8B-Instruct [51]	78.92	43.39	53.56	47.89	55.94	54.20	38.18	50.91	57.50	53.21
GPT-4.1 [37]	75.90	49.00	65.51	32.86	55.82	73.49	40.35	49.74	57.70	56.24
Doubao-Seed-1.6 [45]	80.26	60.00	71.92	35.47	61.91	65.18	47.23	59.49	60.65	60.82
Thinking mode										
Spatial-MLLM-4B [57]	75.27	5.04	10.16	-	30.15	35.87	14.86	29.48	25.78	22.69
Video-R1-7B [14]	74.57	25.50	44.66	4.82	37.39	43.62	26.35	41.66	32.17	31.82
Keye-VL-1.5-8B [62]	73.07	41.61	48.54	38.67	50.47	51.05	30.99	47.59	46.52	45.98
GLM-4.1V-9B [52]	73.81	40.12	55.43	33.12	50.62	50.90	33.19	46.01	45.47	42.46
Intern-VL3.5-14B [68]	78.48	42.96	57.90	24.97	51.08	50.21	39.93	51.13	46.27	47.67
Intern-VL3.5-8B [68]	79.55	43.74	55.10	36.50	53.72	51.23	35.34	51.61	53.74	50.05
MiMo-VL-7B-RL-2508 [50]	65.10	59.62	65.00	35.33	56.26	46.61	38.96	45.38	41.46	50.85
Doubao-Seed-1.6 [45]	76.16	61.94	71.93	38.44	62.12	60.31	46.20	58.05	61.02	63.35
Gemini-2.5-flash [12]	84.16	65.12	68.96	38.04	64.07	79.95	40.21	59.15	63.07	59.46
GPT-5-pro [38]	86.87	67.90	75.89	34.81	66.37	82.85	46.95	61.74	70.58	64.78
Gemini-2.5-pro [12]	84.30	73.00	79.15	32.29	67.18	85.96	54.47	61.21	70.87	65.51

Table 2. CapNav performance across 13 VLMs. Each number indicate the best performance under all tested frame settings. Left block reports each task metrics (Feas-F1, PV, RTA, RV, CapNav). Right block reports per-agent composite scores for the five agent types: adult with no motor disabilities, humanoid robot, quadrupedal, sweeping robots, and wheelchair users. Bold number indicate best per block. Blue indicate open-sourced model; Green indicate proprietary; Orange indicate spatial reasoning models.

4. Experiments

Our experiments address three research questions:

1. **RQ1:** How do SOTA VLMs perform on CAPNAV?
2. **RQ2:** How does performance vary across models, agent types, and inference settings?
3. **RQ3:** What failure modes reveal model limitations?

To answer these questions, we evaluate 13 modern VLMs on CAPNAV. We report quantitative results stratified by model family, agent type, inference setting, and challenge type. We further analyze common failure cases and consolidate them into a qualitative error taxonomy.

4.1. Settings

We evaluated 13 VLMs in November 2025, spanning popular proprietary and open-source models with strong multimodal capabilities, including the Gemini 2.5 family, GPT 4.1 and 5-Pro, Doubao-Seed, and Qwen3-VL. For models with a thinking option, we test under both *thinking* and *non-thinking* configurations. We additionally include Spatial-MLLM [57] and Video-R1 [14], which explicitly target spatial reasoning. Proprietary models are accessed via APIs, while open-source models are deployed on either a single NVIDIA A100 GPU (80GB) or a single A40 GPU (48GB).

At inference time, each model receives CAPNAV tasks as a triple $(\mathcal{S}, \tau, \mathbf{a})$ (Section 3.1). A practical limitation of current VLM interfaces is the maximum number of visual tokens/frames accepted per request. To study this constraint, we evaluate up to four input settings for each model (subject

to interface limits): **16**, **32**, and **64** total frames per clip, and a **1 FPS** setting (avg. 163 frames/video).

4.2. Performance

We compare model outputs against the ground-truth navigation graph to compute overall and per-agent performance. Table 2 summarizes performance metrics across all models under both *thinking* and *non-thinking* settings. We show the best metrics when multiple input frame settings are tested.

All evaluated models outperform the random-walk baseline (CapNav = 29.35%). The strongest performance is achieved by proprietary models (notably the Gemini-2.5 family and GPT-5-pro), which exceed the human average (CapNav = 60.59%) but remain below the best human performance (CapNav = 74.77%). Overall, results show a clear separation between top-tier models and the remainder.

Models explicitly designed for spatial reasoning (Spatial-MLLM and Video-R1-7B) perform substantially worse across all metrics, despite claims of strong spatial performance. This indicates that their spatial reasoning training schemes and architectural modifications are not yet sufficient to address the challenges posed by CAPNAV.

We also test the sensitivity of rankings to the CapNav score composition. In addition to equal weighting, we evaluate four alternative schemes that assign weight 0.5 to one component and 1/6 to each of the remaining components. Kendall’s τ between rankings across these schemes averages 0.909, suggesting that conclusions are largely robust

to moderate weight changes.

4.3. Failure-mode taxonomy

To characterize systematic errors, we use Gemini-3 to mine recurring error themes from 1,000 randomly sampled inference outputs, and then manually refine these themes into a four-category taxonomy. We then visualize the VLM predicted results of 1500 sampled QA pairs to measure the prevalence and distribution of each error type.

Path hallucination: (N = 659 / 1500) the predicted path links non-adjacent nodes, or violates the specified start/end, indicating an incorrect reasoning of the scene connectivity.

Obstacle hallucination: (N = 418) the model reports non-existent blocking obstacles (e.g., closed doors or clutter), suggesting an unreliable interpretation of visual evidence.

Dimension neglect: (N = 191) the model overlooks geometric constraints that block the agent (e.g., narrow doors, tight clearances, cluttered passages), reflecting weak grounding between spatial extent and traversability.

Ability hallucination: (N = 10) the output reasoning contradicts the provided agent profile. This failure is observed in smaller open-sourced models (e.g., MiMo-VL 7B).

4.4. Physical Obstacle Types

We observe four major types of physical obstacles in the ground truth: **stairs** (N = 520 / 2365), which block humanoid robots, sweeping robots, and wheelchair users; **door sill/floor height differences** (N = 82), which stop wheelchair users and sweeping robots; **narrow pathways** (N = 438), usually caused by narrow doors or furniture clutter, which can restrict humanoid robots and wheelchair users; **lack of turning spaces** (N = 28), which hinders wheelchair users and, in rare cases, quadrupedal robots. For the navigation tasks labeled as non-traversable due to one or more of these obstacles, we evaluate how often VLMs can both correctly predict infeasibility and provide appropriate reasoning. The results are visualized in Figure 5.

5. Findings

Drawing on both quantitative results and qualitative inspection of model outputs, we distill three primary findings. Collectively, they indicate that current VLM navigation remains brittle under capability constraints, visually demanding inputs, and geometry-sensitive traversability judgments.

5.1. Capability constraints induce systematic performance degradation

Across all tested models, performance is highest under the baseline embodiment without mobility limitations (adults without mobility constraints; mean CAPNAV score = 57.83%). As constraints tighten, performance degrades. The largest gap appears for the humanoid embodiment due

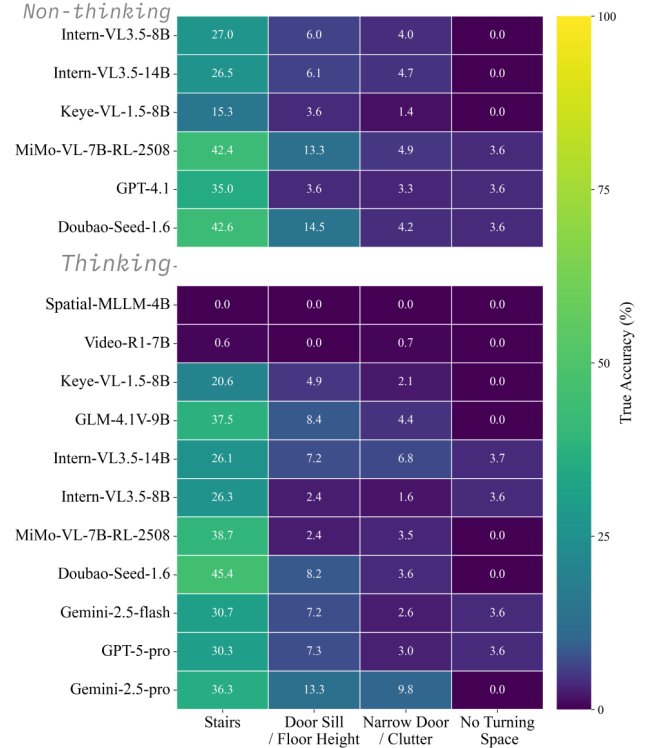


Figure 5. Model accuracy by obstacle type.

to (i) its inability to climb stairs and (ii) a minimum pathway clearance requirement of 0.9 m; this configuration attains the lowest average score (mean = 39.12%).

This pattern highlights systematic deficits and imbalances in VLM-based navigation under mobility constraints. Strong performance in baseline settings does not translate to capability-constrained scenarios; hence, targeted evaluation is essential before deployment in safety-critical navigation.

The five agent types supported in CAPNAV represent common human and robotic mobility profiles. As such, our results generalize to a wide range of practical embodiments. Practitioners may approximate their own systems to the closest of the five provided profiles. For embodiments that differ substantially, we open-source our annotation interface and data curation pipeline to allow new agent profiles. This facilitates capability-aware evaluation for diverse hardware platforms and emerging robotic systems.

5.2. Increasing visual budget helps, but exposes a visual bottleneck

Comparing 9 pairs of inference experiments with and without *thinking*, we find consistent improvements by an average of $\Delta\text{CAPNAV} = 6.87\%$ when *thinking* is enabled. This gain is accompanied by a substantial compute cost: in our measurements, mean inference time increases by approximately $8\times$ (from 14.94 s to 123.94 s).

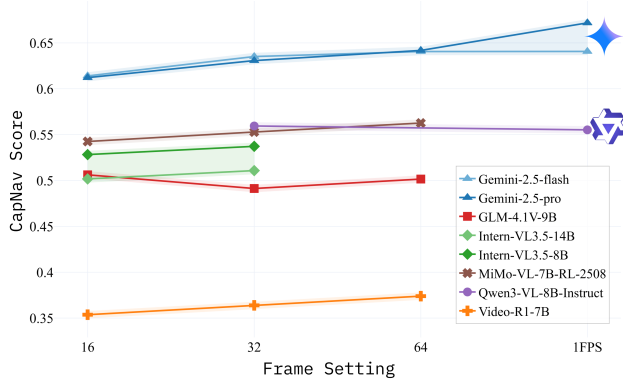


Figure 6. CapNav performance across frame-budget settings.

Across varying frame settings, latency remain stable but higher frame counts increase GPU VRAM and API costs. Performance improves unevenly with more frames, tending to be larger for stronger models (notably Gemini) and negligible for weaker models (see Figure 6). This pattern reveals a *visual bottleneck*: additional evidence only helps if the model can reliably integrate it.

Qualitative inspection suggests two explanations: (i) For most open-sourced models, 64 frames is already a very heavy input, but frames still remain sparse in large scenes, requiring robust cross-frame reasoning that weaker models lack. (ii) Additional viewpoints can increase *obstacle hallucinations*: salient but non-blocking cues may be misinterpreted as traversal barriers, echoing prior observations about limitations in nonlocal reasoning for VLMs [4].

These results imply a cost-benefit frontier rather than a monotonic “more is better” rule. For CAPNAV-like VLN tasks, increasing inference budget can improve performance, but the return depends on model capacity and may trade off against latency, cost, and even error modes.

5.3. Dimension neglect

From our obstacle type analysis in Figure 5, all tested models perform markedly better on obstacles with salient visual signatures (e.g., stairs, door sills) than on constraints requiring implicit metric estimation (e.g., narrow clearances, turning radii). These failures indicate a recurring form of *dimension neglect*: traversability decisions that hinge on estimating size, clearance, or maneuvering space from images remain challenging for all tested VLMs.

To probe whether the above deficits are learnable, we ran a pilot fine-tuning experiment. We augment the CapNav dataset on the navigation graph to expand the task count from 2K to 13K, and LoRA fine-tune on Qwen3-VL-8B-Instruct. After one epoch of training, test score improves from 45.26% to 55.18%, and the finetuned model generates more dimension-aware reasoning. However, reasoning validity decreases (0.30 $\rightarrow$ 0.25), and we observe frequent

false positives on narrow passages (i.e., incorrectly flagging feasible routes as blocked).

These outcomes suggest that fine-tuning can reduce dimension neglect but does not fully resolve geometric precision, and may shift the error profile toward overly aggressive obstacle judgments. This highlights the need to jointly optimize for *obstacle awareness* and *dimension reasoning*. We plan to explore RL-based tuning with task-level rewards under embodiment constraints. We also plan to experiment with spatial-information injection (depth estimates, topological priors, or map-like intermediate representations) to improve dimension judgments.

6. Conclusion

We present CAPNAV, a benchmark that systematically evaluates capability-conditioned navigation across diverse embodiments, realistic mobility obstacles, and graph-structured indoor environments. By evaluating 13 state-of-the-art VLMs, we observe three consistent patterns. First, performance degrades systematically as mobility constraints tighten, indicating that strong results in unconstrained human-like settings do not transfer reliably to other embodiments. Second, increasing the inference budget (more frames or “thinking” mode) yields inconsistent gains and expose a visual integration bottleneck. Third, models exhibit dimension neglect and struggle with dimension reasoning (e.g., narrow clearances and turning radii). A pilot fine-tuning study suggests that these deficits are partially learnable, but fine-tuning may introduce false positives, highlighting the need to jointly optimize the spatial reasoning reliability in task formulation and spatial dimension. We release both the CAPNAV dataset and annotation tool.

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